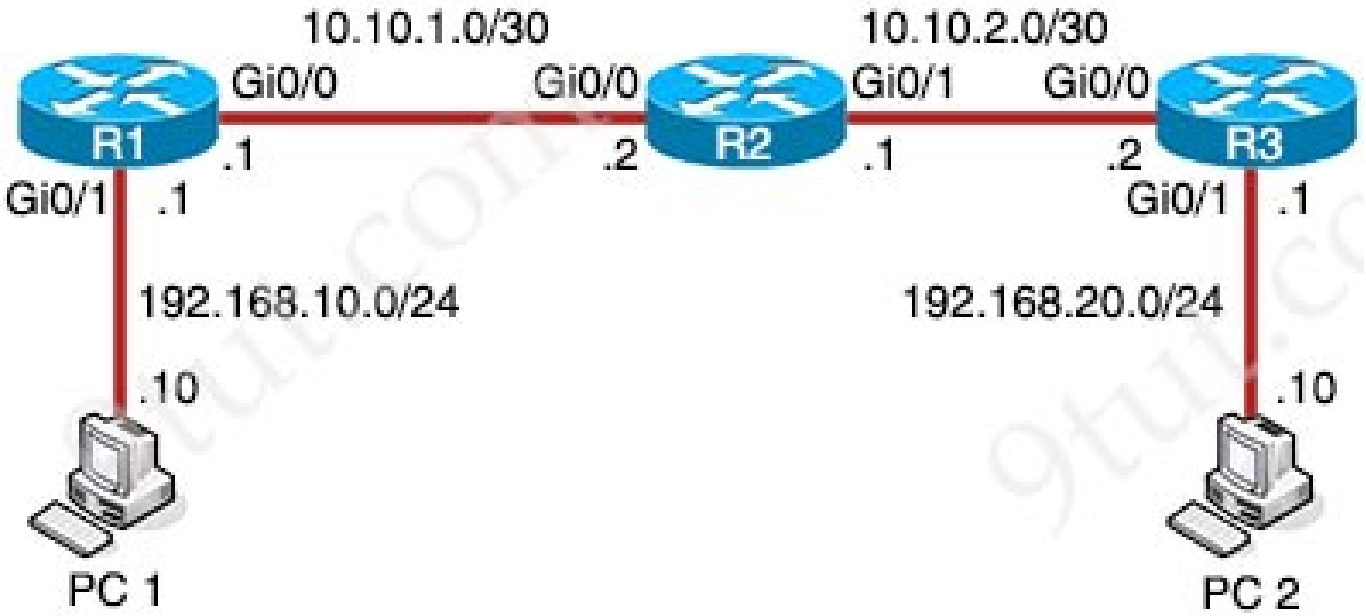
IP Routing Questions

14 questions | Original English questions and source answers

Question 1

Refer to the exhibit. When PC 1 sends a packet to PC2, the packet has which source and destination IP address
when it arrives at interface Gi0/0 on router R2?



Options

A. source 192.168.20.10 and destination 192.168.20.1

B. source 192.168.10.10 and destination 192.168.20.10

C. source 10.10.1.1 and destination 10.10.2.2

D. source 192.168.10.10 and destination 10.10.2.2

Answer

B. source 192.168.10.10 and destination 192.168.20.10

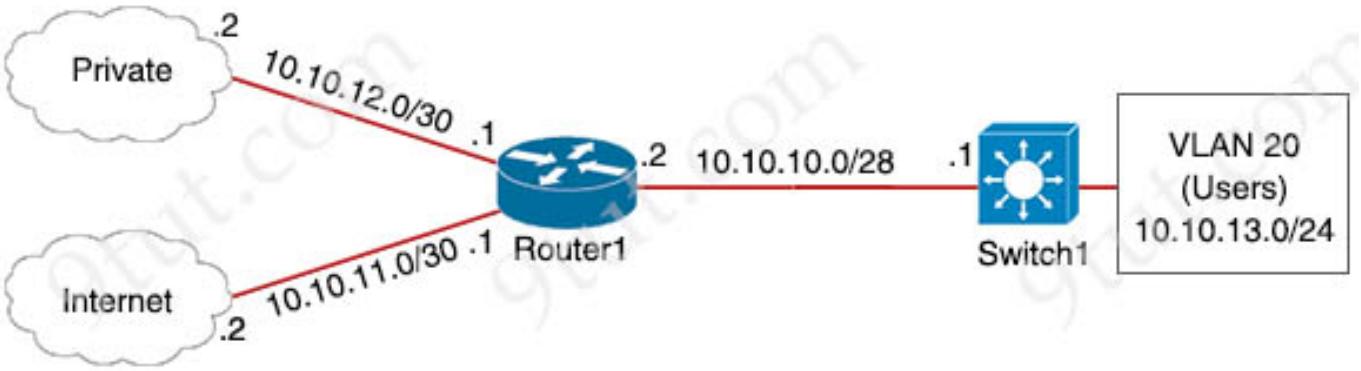
Source explanation

The source and destination IP addresses of the packets are unchanged on all the way. Only source and destination MAC addresses are changed.

Source: CCNA Training » IP Routing Quiz.pdf | Question 1

Question 2

Refer to the exhibit. Which path is used by the router for Internet traffic?
R1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
209.165.200.0/27 is subnetted, 1 subnets
B
209.165.200.224 [20/0] via 10.10.12.2, 00:10:34
10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C
10.10.10.0/28 is directly connected, GigabitEthernet0/0
C
10.10.11.0/30 is directly connected, FastEthernet2/0
C
10.10.13.0/30 [110/2] via 10.10.10.1, 00:03:34, GigabitEthernet0/0
C
10.10.12.0/30 is directly connected, GigabitEthernet0/1
S*
0.0.0.0/0 [1/0] via 10.10.11.2
Switch1#show ip route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C
10.10.10.0/28 is directly connected, FastEthernet0/1
C
10.10.13.0/24 is directly connected, VLAN20



Options

A. 10.10.13.0/24

B. 0.0.0.0/0

C. 209.165.200.0/27

D. 10.10.10.0/28

Answer

B. 0.0.0.0/0

Source: CCNA Training » IP Routing Quiz.pdf | Question 2

Question 3

When a floating static route is configured, which action ensures that the backup route is used when the primary
route falls?

Options

A. The floating static route must have a lower administrative distance than the primary route so it is used as a backup

B. The administrative distance must be higher on the primary route so that the backup route becomes secondary

C. The floating static route must have a higher administrative distance than the primary route so it is used as a backup

D. The default-information originate command must be configured for the route to be installed into the routing table

Answer

C. The floating static route must have a higher administrative distance than the primary route so it is used as a backup

Source: CCNA Training » IP Routing Quiz.pdf | Question 3

Question 4

Refer to the exhibit. Which route does R1 select for traffic that is destined to 192.168.16.2?
R1#show ip route
D 192.168.16.0/26 [90/2679326] via 192.168.1.1
R 192.168.16.0/24 [120/3] via 192.168.1.2
O 192.168.16.0/21 [110/2] via 192.168.1.3
i L1 192.168.16.0/27 [115/30] via 192.168.1.4

Options

A. 192.168.16.0/27

B. 192.168.16.0/26

C. 192.168.16.0/24

D. 192.168.16.0/21

Answer

A. 192.168.16.0/27

Source explanation

The destination IP addresses match all four entries in the routing table but the 192.168.16.0/27 has the longest
prefix so it will be chosen. This is called the “longest prefix match” rule.

Source: CCNA Training » IP Routing Quiz.pdf | Question 4

Question 5

Which statement about static and dynamic routes is true?

Options

A. Static routes tell the router how to forward packets to networks that are not directly connected, while dynamic routes tell the router how to forward packets to networks that are directly connected

B. Dynamic routes are manually configured by a network administrator, while static routes are automatically learned and adjusted by a routing protocol

C. Static routes are manually configured by a network administrator, while dynamic routes are automatically learned and adjusted by a routing protocol

D. Dynamic routes tell the router how to forward packets to networks that are not directly connected, while static routes tell the router how to forward packets to networks that are directly connected

Answer

C. Static routes are manually configured by a network administrator, while dynamic routes are automatically learned and adjusted by a routing protocol

Source: CCNA Training » IP Routing Quiz.pdf | Question 5

Question 6

Which attribute does a router use to select the best path when two or more different routes to the same destination exist from two different routing protocols?

Options

A. dual algorithm

B. hop count

C. administrative distance

D. metric

Answer

C. administrative distance

Source: CCNA Training » IP Routing Quiz.pdf | Question 6

Question 7

Router A learns the same route from two different neighbors, one of the neighbor routers is an OSPF neighbor and the other is an EIGRP neighbor. What is the administrative distance of the route that will be installed in the
routing table?

Options

A. 110

B. 115

C. 90

D. 20

Answer

C. 90

Source explanation

The Administrative distance (AD) of EIGRP is 90 while the AD of OSPF is 110 so EIGRP route will be chosen to install into the routing table.

Source: CCNA Training » IP Routing Quiz.pdf | Question 7

Question 8

Refer to the exhibit. With which metric was the route to host 172.16.0.202 learned?

```
R1#show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
     is directly connected, Serial0/1/0
     172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
S    172.16.0.0/24 [1/0] via 207.165.200.254, 00:00:33, Serial0/0/0
O    172.16.0.128/25 [110/38443] via 207.165.200.254, 00:00:33, Serial0/0/1
D    172.16.0.192/29 [90/3184439] via 207.165.200.254, 00:00:33, Serial0/0/1
     209.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C    209.165.200.248/30 is directly connected, Serial0/0/0
L    209.165.200.249/30 is directly connected, Serial0/0/0
C    209.165.200.252/30 is directly connected, Serial0/0/1
L    209.165.200.253/32 is directly connected, Serial0/0/1
```

Options

A. 0

B. 110

C. 38443

D. 3184439

Answer

C. 38443

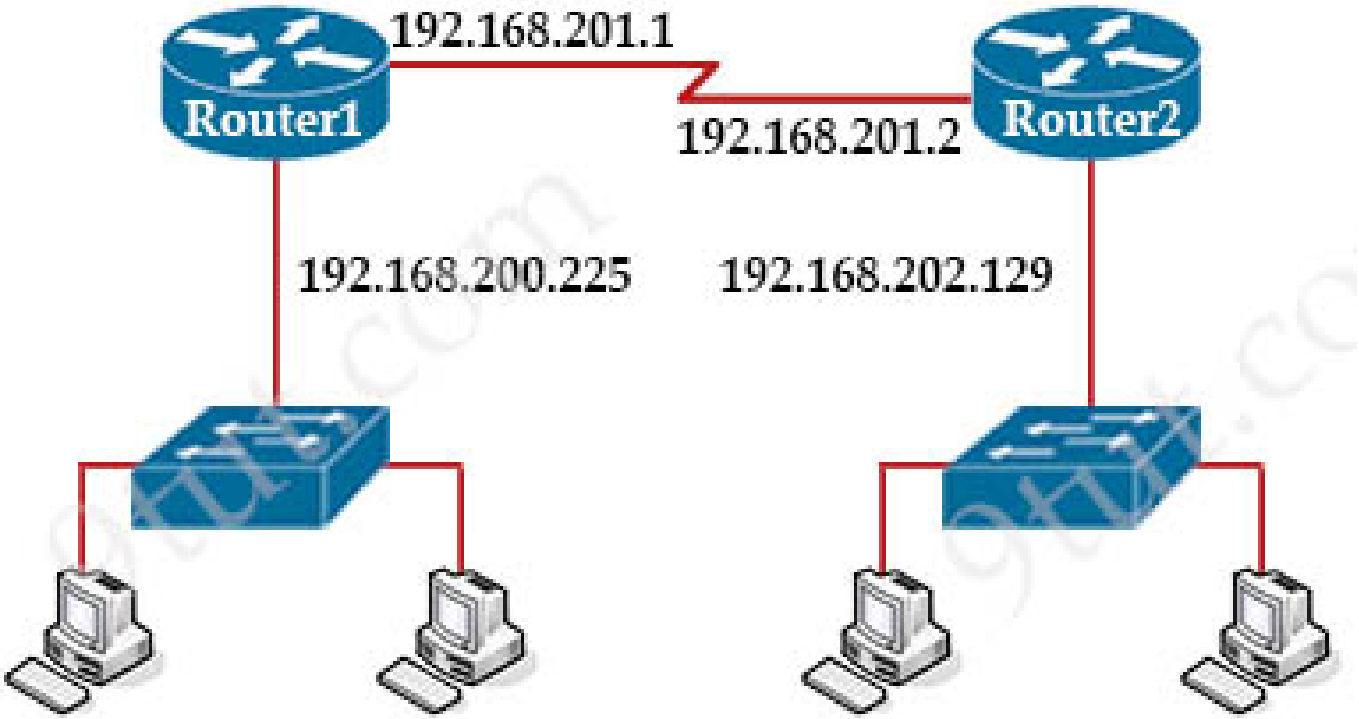
Source explanation

Both the line “O 172.16.0.128/25” and “S 172.16.0.0/24” cover the host 172.16.0.202 but with the “longest (prefix) match” rule the router will choose the first route.
Note: Network 172.16.0.192/29 does not cover host 172.16.0.202. For network 172.16.0.192/29:
Increment: 8
Network address: 172.16.0.192
Broadcast address: 172.16.0.199

Source: CCNA Training » IP Routing Quiz.pdf | Question 8

Question 9

Refer to the exhibit. Which command would you use to configure a static route on Router 1 to network 192.168.202.0/24 with a nondefault administrative distance?



Options

A. router1(config)#ip route 1 192.168.201.1 255.255.255.0 192.168.201.2

B. router1(config)#ip route 192.168.202.0 255.255.255.0 192.168.201.2 5

C. router1(config)#ip route 5 192.168.202.0 255.255.255.0 192.168.201.2

D. router1(config)#ip route 192.168.202.0 255.255.255.0 192.168.201.2 1

Answer

B. router1(config)#ip route 192.168.202.0 255.255.255.0 192.168.201.2 5

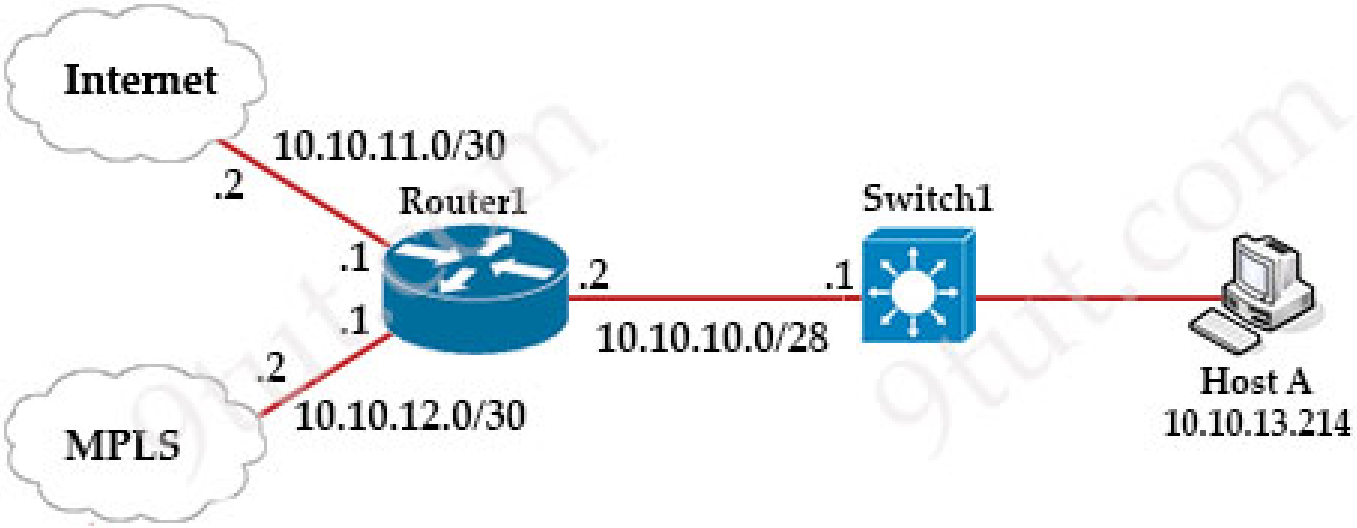
Source explanation

The default AD of static route is 1 so we need to configure another number for the static route.

Source: CCNA Training » IP Routing Quiz.pdf | Question 9

Question 10

Refer to the exhibit. Which prefix does Router 1 use for traffic to Host A?
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
209.165.200.0/27 is sudnetted, 1 subnets
B 209.165.200.224 [20/0] via 10.10.12.2, 03:03:03
209.165.201.0/27 is sudnetted, 1 subnets
B 209.165.201.0 [20/0] via 10.10.12.2, 03:03:03
209.165.202.0/27 is subnetted, 1 subnets
B 209.165.202.128 [20/0] via 10.10.12.2, 03:03:03
10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C 10.10.10.0/28 is directly connected, GigabitEthernet0/0
C 10.10.11.0/30 is directly connected, FastEthernet2/0
C 10.10.12.0/30 is directly connected, GigabitEthernet0/1
O 10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O 10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O 10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O 10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O 10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
S* 0.0.0.0/0 [1/0] via 10.10.11.2



Options

A. 10.10.10.0/28

B. 10.10.13.208/29

C. 10.10.13.0/25

D. 10.10.13.144/28

Answer

B. 10.10.13.208/29

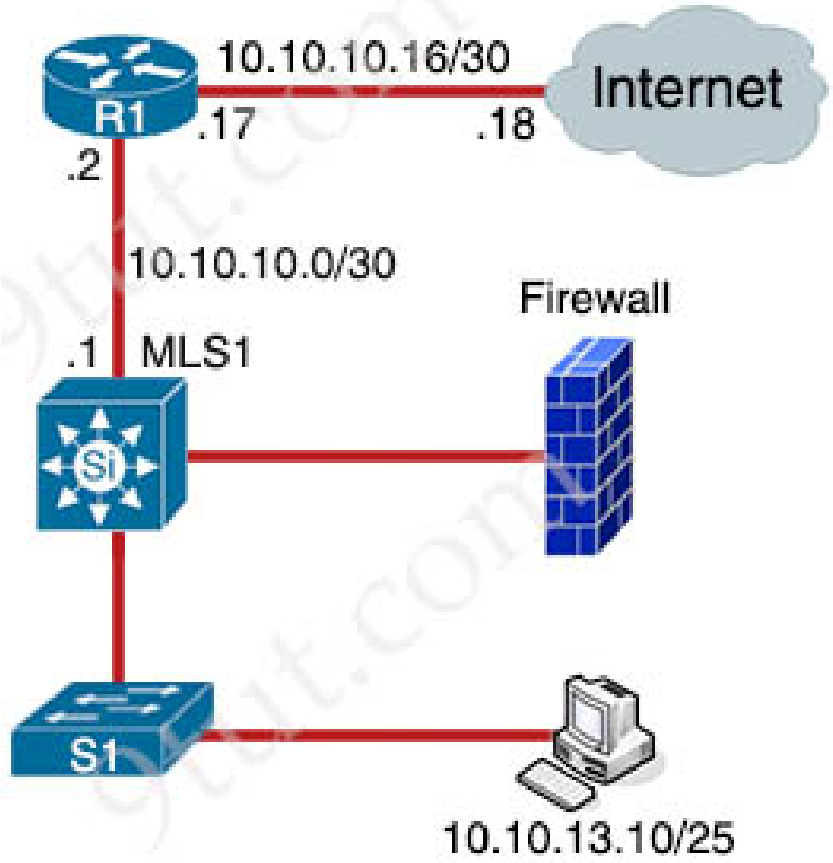
Source explanation

The prefix with “longest prefix” will be matched first, in this case is “/29”.

Source: CCNA Training » IP Routing Quiz.pdf | Question 10

Question 11

Refer to the exhibit. Which type of route does R1 use to reach host 10.10.13.10/32?
R1#show ip route
Gateway of last resort is 10.10.10.18 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 4 subnets. 3 Masks
C 10.10.10.0/30 is directly connected. FastEthernet0/1
O 10.10.13.0/25 [110/4576] via 10.10.10.1, 02:53:11, FastEthernet0/1
C 10.10.10.16/30 is directly connected, FastEthernet0/24
0 10.10.13.144/28 [110/110] via 10.10.10.1, 03:51:21, FastEthernet0/1
B* 0.0.0.0/0 [20/0] via 10.10.10.18, 02:17:53



Options

A. network route

B. floating static route

C. default route

D. host route

Answer

A. network route

Source explanation

From the output, we see R1 will use the entry “O 10.10.13.0/25 [110/4576] via 10.10.10.1, …” to reach host 10.10.13.10. This is a network route.
Note: “B* 0.0.0.0/0 …” is a default route.

Source: CCNA Training » IP Routing Quiz.pdf | Question 11

Question 12

Refer to the exhibit. How does router R1 handle traffic to 192.168.10.16?
R1# show ip route
D 192.168.10.0/24 [90/2679326] via 192.168.1.1
R 192.168.10.0/27 [120/3] via 192.168.1.2
O 192.168.10.0/28 [110/2] via 192.168.1.3
i L1 192.168.10.0/13 [115/30] via 192.168.1.4

Options

A. It selects the OSPF route because it has the lowest cost

B. It selects the IS-IS route because it has the shortest prefix inclusive of the destination address

C. It selects the RIP route because it has the longest prefix: inclusive of the destination address

D. It selects the EIGRP route because it has the lowest administrative distance

Answer

C. It selects the RIP route because it has the longest prefix: inclusive of the destination address

Source explanation

Note: The "O 192.168.10.0/28" route ranges from 192.168.10.0 to 192.168.10.15 only so it does not include destination 192.168.10.16 so OSPF route is not the correct answer.

Source: CCNA Training » IP Routing Quiz.pdf | Question 12

Question 13

Refer to the exhibit. If configuring a static default route on the router with the ip route 0.0.0.0 0.0.0.0 10.13.0.1
120 command, how does the router respond?
Gateway of last resort is 10.12.0.1 to network 0.0.0.0
O*E2 0.0.0.0/0 [110/1] via 10.12.0.1, 00:00:01, GigabitEthernet0/0
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/24 is directly connected, GigabitEthernet0/0
L 10.0.0.2/32 is directly connected, GigabitEthernet0/0
C 10.13.0.0/24 is directly connected, GigabitEthernet0/1
L 10.13.0.2/32 is directly connected, GigabitEthernet0/1

Options

A. It starts sending traffic without a specific matching entry in the routing table to GigabitEthernet0/1

B. It immediately replaces the existing OSPF route in the routing table with the newly configured static route

C. It ignores the new static route until the existing OSPF default route is removed

D. It starts load-balancing traffic between the two default routes

Answer

C. It ignores the new static route until the existing OSPF default route is removed

Source explanation

Our new static default route has the Administrative Distance (AD) of 120, which is bigger than the AD of OSPF External route (O*E2) so it will not be pushed into the routing table until the current OSPF External route is removed.
For your information, if you don’t type the AD of 120 (using the command “ip route 0.0.0.0 0.0.0.0 10.13.0.1”) then the new static default route would replace the OSPF default route as the default AD of static route is 1. You
will see such line in the routing table:
S* 0.0.0.0/0 [1/0] via 10.13.0.1

Source: CCNA Training » IP Routing Quiz.pdf | Question 13

Question 14

Which of the following dynamic routing protocols are Distance Vector routing protocols? (Choose two)

Options

A. OSPF

B. BGP

C. RIP

D. IS-IS

E. EIGRP

Answer

C. RIP

E. EIGRP

Source: CCNA Training » IP Routing Quiz.pdf | Question 14